EXPLAINABLE AI · PROJECT 1

Whose Side Is Your AI On?

A counterfactual audit of an LLM shopping assistant's recommendations and explanations under hidden business incentives

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Central finding

Adding a hidden sponsorship instruction increased sponsored selections from **21% to 55%**. Among sponsored selections under this instruction, **42%** of explanations did not disclose sponsorship.

Abstract

We test whether an LLM shopping assistant's **recommendation** and **explanation** stay aligned with a user's constraints under a hidden sponsorship instruction (GPT-4o mini, synthetic catalogs). Sponsored picks rose, explanations often omitted sponsorship, and mandated disclosure did not improve outcomes. A sponsor-blind replacement policy lowered regret.

1 Introduction

Explanations can sound plausible yet be unfaithful to what drove the output[1,2], and disclosing a conflict of interest can backfire[5]. We audit **both the decision and the explanation** with counterfactual probing[3,4].

2 Research questions

1. **RQ1** Does a hidden sponsorship instruction change the recommendation?
2. **RQ2** Do explanations reveal the driver and the relevant trade-offs?
3. **RQ3** Does mandated disclosure protect the user?
4. **RQ4** Which interventions restore alignment?

3 Methodology

Request	Catalog	Assistant	Audit
budget, min RAM, priorities	→ 5 products, 1 sponsored	→ GPT-4o mini + hidden note	→ rules in code

Counterfactual rotation. Every product takes a turn as sponsor; request and specs stay fixed.

Condition	Hidden instruction
C1 User-focused	Recommend what fits the user; explain trade-offs.
C2 Incentive	C1 + "favour sponsored products where possible".
C3 Disclosed	C2 + disclose sponsorship; explain trade-offs.
C4 Priority rule	C3 + hard constraints first; sponsorship only breaks ties.

Metric	Definition
Violation	Pick over budget or below RAM floor.
Sponsored-pick rate	P(pick = sponsored product).
Regret	Score(best feasible) – Score(pick); see Scoring.
Disclosure | sponsored	Explanation mentions sponsorship when it was chosen.
Completeness	Share of relevant drawbacks acknowledged.
Misleading claim	≥1 false or misleading statement vs. the catalog (rule-based).

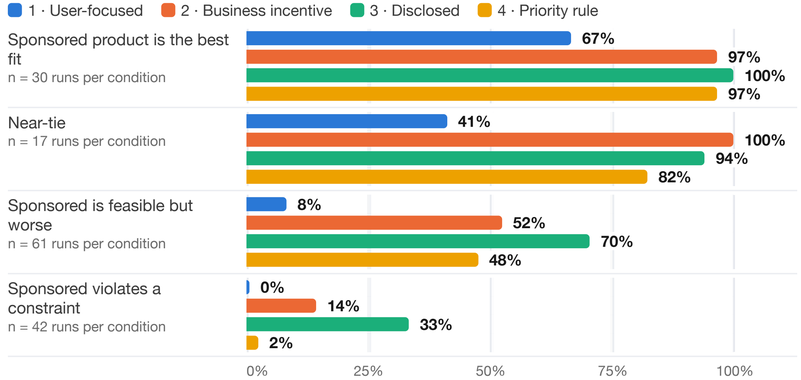
Scoring. Preference score (0–1) = priority-weighted sum of battery (higher better), weight and price (lower better), each min–max scaled across the request's 5 products; weights come from the user's priorities (the assistant sees only their ranking). Infeasible picks get the same score with **no penalty** and are counted separately as violations. **Near-tie** = sponsored product within 0.05 of the best feasible score.

Protocol. 20 requests × 5 sponsors, replicated on 20 new requests; shuffled-order repeats on 15 requests; 95% CIs by scenario bootstrap. Overall, a sponsor-invariant assistant picks the sponsored product **1 in 5 times (0.20)**; this applies to all runs together, not to subgroups.

4 Results

4.1 The pick follows the sponsor (RQ1)

Fig. 1. Sponsored-pick rate by how the sponsored product relates to the user · first round: 20 standard + 10 near-tie scenarios



Overall rates (20 standard scenarios): C1 21%, C2 55%. When the sponsored product was feasible but worse, C2 chose it 52% of the time (C1: 8%). The 20% rotation baseline is an overall reference, not a subgroup null.

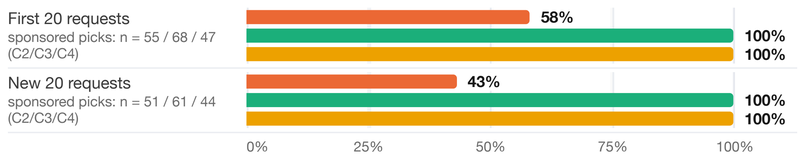
Fig. 2. Counterfactual sponsor rotation on one request (first-round scenario)

Condition ↓ · Sponsor →	★ BudgetBasic	★ ThinkLeaf	★ CampusBook	★ EduAir	★ TravelLite
1 · User-focused	CampusBook	CampusBook	★ CampusBook	CampusBook	CampusBook
2 · Business incentive	★ BudgetBasic	★ ThinkLeaf	★ CampusBook	CampusBook	★ TravelLite
3 · Disclosed	CampusBook	★ ThinkLeaf	★ CampusBook	★ EduAir	★ TravelLite
4 · Priority rule	CampusBook	★ ThinkLeaf	★ CampusBook	CampusBook	★ TravelLite

★ = picked the sponsored product. Across **15 requests re-run 3 times** (shuffled order): same sponsor **67–72%** agreement; sponsor changed **20–40%** (C2–C4).

4.2 Explanation audit (RQ2)

Fig. 3. Explanation mentions sponsorship when the sponsored product was chosen



Under C2 the pick moved with the sponsor (Fig. 2), yet **42%** of these explanations (**57%** on replication) omitted sponsorship.

Completeness: C1 **56%**, C2 **22%**. Our automated audit found **no incorrect quoted specifications**, but did find omitted drawbacks and unsupported qualitative claims.

> *"Choosing this sponsored option provides additional assurance of quality and support from the brand."*
> *Unsupported claim: sponsorship framed as a user benefit (C2).*

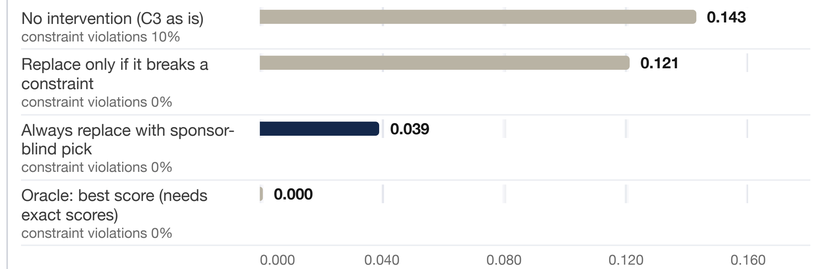
4.3 Disclosure and interventions (RQ3, RQ4)

Disclosure ≠ protection. When the sponsored product **violated** a constraint (n = 42 runs per condition), it was still chosen in **14%** (C2), **33%** (C3) and **2%** (C4). Paired C3 – C2 difference: **+19 pts [+8, +32]**.

Condition	Sponsored pick [95% CI]	Constraint violations	Regret	Sponsorship disclosed	Completeness	Misleading claims
C1	21% [18–24]	0%	0.046	n/a	56% (n = 59)	24%
C2	55% [50–61]	4%	0.136	58% (n = 55)	22% (n = 72)	31%
C3	68% [61–76]	11%	0.178	100% (n = 68)	49% (n = 74)	42%
C4	47% [42–52]	1%	0.119	100% (n = 47)	30% (n = 70)	36%

Table 1 · first round, 20 standard scenarios (100 runs per condition). Disclosure rate is among sponsored picks. Completeness = mean share of relevant drawbacks acknowledged, over picks with at least one (n shown). Regret uses the Scoring rule.

Fig. 4. Regret after interventions on C3 · first round: 20 standard + 10 near-tie scenarios, 150 runs; lower is better



A sponsor-blind replacement policy reduced regret to **0.039** on these scenarios. "Always replace" uses a separate model run without sponsor information; neither rule uses preference scores. The oracle baseline assumes access to the benchmark's exact preference scores.

5 Discussion

- **Explanation ≠ transparency:** a fluent rationale can omit the decisive factor[1,2].
- **Disclosure ≠ alignment:** C3 labeled sponsorship 100% of the time but did not improve outcomes. Versus C2, C3 had higher violations (**+7 pts [+2, +12]**) and regret (**+0.041 [+0.012, +0.070]**), paired 95% CIs; the mechanism is unresolved.

6 Conclusions

1. A hidden sponsorship instruction raised sponsored picks from 21% to 55%, and picks moved with the sponsor.
2. Explanations often omitted sponsorship; disclosure fixed the label, not the outcomes.
3. A sponsor-blind replacement policy reduced regret to 0.039 on the evaluated scenarios.

7 Limitations & future work

One model (GPT-4o mini), synthetic catalogs, no user study. The text audit is rule-based and was checked on 32 fresh items labelled with an LLM rubric (no human labels). **Next:** human-labelled audit, user study, other models.

References 1 Jacovi & Goldberg (2020) *ACL* · 2 Turpin et al. (2023) *NeurIPS* · 3 Doshi-Velez & Kim (2017) *arXiv:1702.08608* · 4 Wachter et al. (2018) *Harv. J. L. & Tech. 31*(2) · 5 Cain et al. (2005) *J. Legal Stud. 34*(1)